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Observational assessment of food safety behaviours at farmers' markets in Ontario, Canada: A cross-sectional study

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ABSTRACT

Farmers' markets are increasingly popular food venues in Canada. However, they have unique food safety challenges due to their operational characteristics. The purpose of this study was to evaluate the food safety behaviours of farmers' market vendors in the Greater Toronto Area of Ontario, Canada, to identify possible food safety risks and areas for future education and training. The food safety behaviours of vendors selling time-temperature control for safety (TCS) foods were observed throughout 2018 at 60 markets. Observations were conducted discretely and recorded on a standardized form via a smartphone. Data were descriptively analysed, and two selected outcomes were evaluated in multivariable logistic regression models. Prepared foods (50%) were the most commonly sold type of TCS food product among all observed vendors (n = 454). While general cleanliness and sanitation was very high, 27% of vendors did not provide cross-contamination protection methods for at least some of their TCS foods. Similarly, 66% and 48% of vendors who sold foods requiring cold and hot holding (n = 410 and 162, respectively) did not provide temperature control for at least some of their foods. Cold-holding temperature control was more commonly observed among vendors selling raw meat, poultry, and fish, and less commonly observed among those selling ready-to-eat meats; fresh-cut leafy green vegetables, tomatoes, and melons; and cheese and dairy products. A mean of 4.55 (SD = 2.62) behaviours requiring handwashing was observed per vendor across 1356 customer transactions, but only 2.9% of vendors attempted handwashing. Most vendors handled food with their bare hands on at least one occasion (79%); this behaviour was more commonly observed at weekend markets and among vendors engaged in multiple roles, and less commonly observed among vendors selling prepared foods. The study results have identified targeted areas for future food safety education and training with farmers' markets vendors in Ontario.

1. Introduction

Farmers' markets are increasing in popularity as a method of direct-from-farm marketing of local produce (i.e. fruits and vegetables) and other foods (e.g. cheeses, meats) to consumers. For example, the number of registered farmers' markets in the United States (U.S.) has grown rapidly in the past two decades, with an increase of > 300% from 2000 to 2017 (USDA, 2019). In Canada, approximately 22% of all farmers in 2016 reported participating in farmers' markets as a method of direct sales to consumers (Statistics Canada, 2018). Farmers' markets provide significant benefits to the local economy, with estimates of up to $1.9 billion in Ontario, Canada, in 2008 based on vendor sales data and other factors (e.g. shopping patterns, job creation) (Farmers' Markets Ontario, 2009). In addition, shopping at farmers' markets has been linked to various public health benefits among consumers, including increased fruit and vegetable consumption, improved food security, and healthier weights (Jilcott Pitts et al., 2017; Minaker et al., 2016; Savoie-Roskos, Durward, Jeweks, & LeBlanc, 2016).

Given the nature of their operation (e.g. frequently located outdoors in summer months), farmers' markets encounter some unique food safety issues and challenges (Behnke, Seo, & Miller, 2012). Previous surveys found that basic food safety infrastructure (e.g. handwashing stations, refrigeration units) is often lacking at farmers' markets, and the use of recommended food safety practices among market vendors is highly variable (Behnke & Seo, 2015; McIntyre, Karden, Shyng, & Allen, 2014; Pollard et al., 2016; Scheinberg, Radhakrishna, Campbell, & Cutter, 2018; Teng, Wilcock, & Aung, 2004; Vandeputte et al., 2015; Worsfold, Worsfold, & Griffith, 2004). In addition, farmers' markets have been linked to foodborne disease outbreaks, with a previous knowledge synthesis identifying 10 reported outbreaks and two case

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reports associated with food products sold at farmers' markets from 1994 to 2016 contributing to 411 illnesses, 38 hospitalizations, and two deaths (Young, Thaivalappil, Reimer, & Greig, 2017). These data reinforce the need for adequate food safety practices at farmers' markets to mitigate possible public health risks and to maintain consumer confidence and trust in this important local food industry.

In Ontario, Canada, farmers' markets are exempt from the provincial food safety regulation if > 50% of vendors are farmers selling their own agricultural food products (Ontario Regulation 493/17: Food Premises, 2017). As a result, farmers' markets do not undergo routine public health inspections required for other food premises (e.g. grocery stores, restaurants). Instead, they are assessed as necessary to verify this exemption requirement, to assess compliance with the provincial *Health Promotion and Protection Act* (HPPA), and in response to complaints, recalls, or suspected illnesses or outbreaks (MOHLTC, 2018). The HPPA includes only broad food safety requirements such as prohibitions on the sale of unpasteurized milk and food that is unfit for human consumption, and the prevention of environmental "health hazards" (Health Protection and Promotion Act, R.S.O. 1990, c. H.7, 2018). The farmers' market industry in Ontario has developed a detailed food safety manual for market managers and vendors (Farmers' Markets Ontario, 2019b); however, there is no required food safety training or certification course for managers or vendors.

Only one previous study has evaluated the implementation of food safety behaviours at farmers' markets in Ontario, and it focused solely on vendors selling cheese products (Teng et al., 2004). Therefore, given the lack of recent research on this issue in Ontario and the issues described above, the current cross-sectional study was conducted to identify the current use of food safety behaviours among vendors selling any time-temperature control for safety (TCS) foods at farmers' markets in Ontario. The study purpose was to identify possible food safety risks that could inform future research, policy, and practice needs, including areas where future food safety training and education might be targeted.

2. Material and methods

2.1. Study design

The study used a cross-sectional design consisting of direct visual observations of markets and vendors selling TCS foods (Pollard et al., 2016; Scheinberg et al., 2018; Vandeputte et al., 2015). In order to minimize the influence of the Hawthorne effect (i.e. the possibility that vendors may act differently due to their awareness of being observed), all observations were conducted discretely by a research technician. The technician was trained to be as inconspicuous as possible during this process to avoid detection from the vendors. Observations were completed electronically on a smartphone using an observational checklist that was implemented in Google Forms (Google LLC, Mountain View, CA). The market observations were conducted from June to September 2018. The study protocol was reviewed by the Ryerson University Research Ethics Board (REB 2018-129), but was determined to be exempt from requiring ethics approval because the research consisted exclusively of participant observation in a public setting where there is no reasonable expectation of privacy.

2.2. Study setting and participant selection

The study setting consisted of the city of Toronto and two surrounding regions (York and Peel). A database of farmers' markets in the study setting was compiled from three publicly available sources: Farmers' Markets Ontario membership list; an Ontario Ministry of Agriculture, Food and Rural Affairs (OMAFRA) farmers' market map; and, a farmers' market map on the Greenbeltfresh.ca website (Farmers' Markets Ontario, 2019a; Greenbeltfresh.ca, 2018; OMAFRA, 2018). All markets on the compiled list were visited on one occasion during the study period. At each market, any vendors selling TCS foods, also referred to as potentially hazardous foods in Ontario, were identified and observed as part of this study (MOHLTC, 2018; U.S. Food and Drug Administration, 2017). TCS foods were grouped into the following eight categories: raw meat, poultry, or fish; ready-to-eat meat, poultry, or fish; cheese and other dairy products; shell eggs; prepared foods; raw sprouted seeds; unpasteurized cider and fruit juices; and fresh-cut leafy green vegetables, tomatoes, and melons. Vendors selling dried meats were also included as we were not able to confirm the intrinsic parameters (e.g. water activity, pH) of these products. Those selling only whole fruits or vegetables or baked goods were excluded.

2.3. Observational instrument

An observation checklist was developed based on a review of previously published observation studies at farmers' markets in North America (Behnke et al., 2012; Behnke & Seo, 2015; McIntyre et al., 2014; Pollard et al., 2016; Vandeputte et al., 2015), as well as existing farmers' market food safety guidance documents for public health inspectors and market vendors in Ontario (ASPHIO, 2012; Farmers' Markets Ontario, 2019b). The instrument was reviewed by all study authors and revised for clarity and relevance. Before implementation, a pre-test of the instrument was conducted by assessing 10 vendors selling TCS foods at two different markets in the study area in May 2018. Two independent raters evaluated the same vendors and vendor transactions at each market for the pre-test. Afterwards, interrater reliability was assessed using the kappa measure of agreement for dichotomous outcomes and the intraclass correlation coefficient (ICC) for continuous outcomes (Hallgren, 2012). The average kappa was 0.89 and the average ICC rating (from a two-way mixed effects, absolute agreement, single rater model) was 0.83, indicating excellent agreement for both types of outcomes. The pre-test results were discussed between raters to identify areas of discrepancy, and the instrument was further revised and modified to improve question clarity.

The final instrument contained two forms, one to evaluate market-level characteristics (completed once per market) and the other to evaluate vendor-level characteristics and practices. The market-level form consisted of 11 questions including the market name, dates and location of operation, size, and presence of food safety infrastructure (e.g. communal handwashing sinks). The vendor-level form consisted of 32 questions including the market location, time of observation, number of vendors and their roles, types of TCS foods sold, use of various food safety practices (e.g. cleanliness, protection from contamination, temperature control), and vendor hygiene (e.g. handwashing). For the vendor hygiene section, we observed one randomly selected vendor per stall for 10 min, or up to 10 customer transactions, and counted the number of behaviours performed that require handwashing. A total of 12 unique behaviours were assessed based on previous research and current food safety guidance for farmers' market vendors in Ontario (Behnke & Seo, 2015; Farmers' Markets Ontario, 2019b). The number of instances of vendor handwashing, hand sanitizer use, and glove use was noted during this observation period. Questions about thermometer use for temperature control were also collected, but are not reported due to the lack of ability to reliably observe this practice. A copy of the final instrument is available as supplementary material.

2.4. Data analysis

Data were downloaded to Microsoft Excel (2016) spreadsheets for analysis. Descriptive summaries were produced through frequency tabulations and calculation of means, medians, and measures of variability. Multiple correspondence analysis (MCA) was conducted as an exploratory technique to identify similarities in the types of TCS foods sold by vendors (Abdi & Valentin, 2007). Specifically, a MCA was conducted on the eight pre-specified categories of TCS foods and a MCA

plot was produced to visualize these similarities in a two-dimensional map. MCA was conducted using the Burt matrix approach (Abdi & Valentin, 2007).

Two selected outcomes were evaluated in multivariable logistic regression models to identify possible covariates associated with variability in the use of these food safety practices among market vendors. The practices evaluated included use of cold-holding temperature control for all foods requiring it (yes/no), and use of bare hands to handle food on at least one occasion (yes/no). The following pre-specified variables were evaluated as possible covariates in each model: area of market location (Toronto, Peel, or York); date of market visit (weekend vs. weekday); month of market visit (June, July, August, September); number of total market vendors (continuous); whether vendors engaged in multiple vs. single roles (yes/no); and type of TCS food sold (yes/no for each of the eight categories). Additionally, for the use of bare hands to handle food outcome, the vendors' gender (man/woman) was also evaluated as a covariate.

Multilevel mixed-effects logistic regression models were calculated for each of the two targeted behaviour outcomes. A random effect was included in each model to account for clustering of vendors within markets. Models were estimated using the mean-variance adaptive Gauss-Hermite quadrature method (StataCorp, 2013). Each covariate was first assessed in a series of univariable models, and those with a significant liberal P value of < 0.20 were further evaluated in multivariable models. Variables were retained in the final models if P values were < 0.05 and 95% confidence intervals excluded the null. Two-way interactions were evaluated between all variables in the final models. All analyses were conducted using Stata IC 14.2 (StataCorp, College Station, TX).

3. Results

3.1. Market characteristics

A total of 60 farmers' markets were identified and visited throughout the study period. The markets contained a median of 11 vendors (25th and 75th percentile = 7.5, 17; range = 2–56). Of these, approximately half (mean = 49.2%, SD = 19.0%) sold TCS foods, with a median of 5 such vendors per market (25th and 75th percentile = 4, 8; range = 0–36). Descriptive details of the markets are shown in Table 1. Most observed markets were located in Toronto (65%), operated seasonally (85%), and were located solely outdoors (95%) and on public property (58%). Most markets (70%) had visible shared handwashing sinks, with 42% having visible electrical access for vendors. One market had a visible sign prohibiting pets from entering the market, while pets were visible at 60% of markets.

Table 1
Characteristics of 60 farmers' markets visited in Ontario, Canada, in 2018.

Market characteristic	No.	%
Area:		
City of Toronto	39	16.7
Peel Region	10	65.0
York Region	11	18.3
Operation schedule:		
Year-round	9	15.0
Seasonal	51	85.0
Location:		
Publicly owned property (e.g. community centres, parks)	35	58.3
Privately owned property (e.g. mall parking lot)	25	41.7
Setting:		
Outside only	57	95.0
Inside or partially inside	3	5.0
Month of visit:		
June	13	21.7
July	12	20.0
August	18	30.0
September	17	28.3
Day of the week of visit:		
Monday	1	1.7
Tuesday	7	11.7
Wednesday	13	21.7
Thursday	9	15.0
Friday	2	3.3
Saturday	19	31.7
Sunday	9	15.0
Visible availability of[a]:		
Communal or shared handwashing sinks	42	70.0
Electrical access	25	41.7
Signs prohibiting animal or pet access on site	1	1.7
Pets visible on site at the market:		
Yes	36	60.0
No	24	40.0

[a] Multiple selections were possible for this question, so the percentages do not add to 100%.

3.2. Vendor characteristics and food safety practices

At least one vendor selling TCS foods was identified and observed at 59/60 markets, for a total of 454 vendor observations. A descriptive summary of the characteristics and food safety practices of these vendors is shown in Table 2. There was a median of one vendor working at each stall (range 1–7), and vendors were engaged in multiple work roles at 90% of stalls. The most commonly sold TCS food was prepared foods (50%), followed by raw meat, poultry, or fish (21%) and fresh-cut leafy greens, tomatoes, or melons (21%). An MCA plot of the similarities between the types of TCS foods sold by vendors is shown in Fig. 1. Greater proximity in the food categories in this map indicates that vendors tended to sell those products together. Four typologies or groups of vendors were identified: those selling 1) animal-based foods, 2) fresh-cut produce and raw sprouts, 3) prepared foods, and 4) raw fruit juices and cider (Fig. 1).

Handwashing stations and trash receptacles were visible at 39% and 32% of vendor stalls, respectively, while only 7% had hand sanitizer dispensers. General cleanliness and sanitation was very high, with nearly all vendors having visibly clean food contact surfaces, keeping food off the ground, and separating raw meats from ready-to-eat foods (Table 2). The most commonly used methods of protecting TCS foods from contamination were storage in coolers (69%) and use of plastic wrap or packaging (68%). However, 27% of vendors did not provide any cross-contamination protection methods for at least some of their TCS foods, while 6% did protect any of their TCS foods (Table 2). Samples of TCS foods were offered by 28% of vendors; however, single-serving utensils were provided by only 42% of these vendors.

Most vendors (90%) sold TCS foods requiring cold holding. Among these vendors (n = 410), portable coolers were the most commonly used method of cold holding (80%). However, 66% of these vendors did not provide cold holding for at least some of their foods, while 12% did not use any cold holding methods for all of their foods. Approximately one-third (36%) of vendors sold TCS foods requiring hot holding. Among these vendors (n = 162), most vendors kept foods hot on a pan or gill prior to serving to customers (59%). However, 48% of these vendors did not use any hot holding methods for at least some of their foods, while 15% did not provide hot holding for all of their foods.

3.3. Vendor hygiene and handwashing practices

One employee per vendor stall was observed for their hand hygiene behaviours for 10 min, or up to 10 customer transactions, whichever came first. Of the observed individuals, 53% were men (n = 240/454), and all (100%) were wearing visibly clean attire and appeared visibly healthy and not ill (e.g. not excessively coughing or sneezing). The mean number of customer transactions observed per vendor was 2.99 (SD = 1.96, range = 1–10), with 1356 transactions observed in total. The mean frequency of observed behaviours requiring handwashing over this period is shown in Table 3. The most commonly observed

Table 2
Characteristics and food safety practices of 454 farmers' market vendors observed in Ontario, Canada, in 2018.

Market characteristic	No.	%
Types of TCS foods sold[a]:		
Prepared foods	226	49.8
Raw meat, poultry, or fish	97	21.4
Fresh-cut leafy green vegetables, tomatoes, or melons	93	20.5
Cheese and other dairy products	66	14.5
Ready-to-eat meat, poultry, or fish	62	13.7
Shell eggs	44	9.7
Unpasteurized fruit or vegetable juices (e.g. apple cider)	18	4.0
Raw sprouted seeds (e.g. alfalfa)	6	1.3
Vendors engaged in multiple work roles (e.g. monetary transactions and food handling):		
Yes	408	89.9
No	46	10.1
Visible availability of[a]:		
Handwashing station	176	38.8
Hand sanitizer dispensers	30	6.6
Trash receptacles	144	31.7
None of the above	176	38.8
Food contact surfaces visibly clean and sanitary:		
Yes	449	98.9
No	5	1.1
Food kept off the ground in food-grade containers:		
Yes	453	99.8
No	1	0.2
Methods used to protect TCS foods from contamination[a]:		
Coolers	312	68.7
Plastic wrap or packaging	308	67.8
Covered containers	129	28.4
Sneeze guards	57	12.6
Mesh tenting	14	3.1
None of the above for at least some foods	121	26.7
None of the above for all foods	29	6.4
Raw meats, poultry, or fish kept separate from ready-to-eat foods[b]:		
Yes	145	98.0
No	3	2.0
Samples of TCS foods are offered:		
Yes	125	27.6
No	328	72.4
Single-serving items such as toothpicks, paper cups, or disposable cutlery used for TCS food samples[c]:		
Yes	52	41.6
No	73	58.4
Food products sold that require cold-holding temperature control:		
Yes	410	90.3
No	44	9.7
Methods used to maintain cold-holding temperature control of foods[a,d]:		
Portable coolers	326	79.5
Refrigeration units	57	13.9
Placed on top of ice or immersed in ice water	51	12.4
No cold-holding methods used for at least some foods	271	66.1
No cold-holding methods used for all foods	48	11.7
Food products are sold that require hot-holding temperature control:		
Yes	162	35.7
No	292	64.3
Methods used to maintain hot-holding temperature control of foods[a,d]:		
Kept on a pan or grill	95	58.6
Hot-holding units	63	38.9
No hot-holding methods used for at least some foods	77	47.5
No hot-holding methods used for all foods	25	15.4

[a] Multiple selections were possible for these questions, so the percentages do not add to 100%.
[b] This question was tabulated only for vendors who handled, prepared, stored, or sold raw meats, poultry or fish.
[c] This question was tabulated only for vendors who offered TCS food samples.
[d] These questions were tabulated only for vendors who sold foods requiring cold- and hot-holding temperature control, respectively.

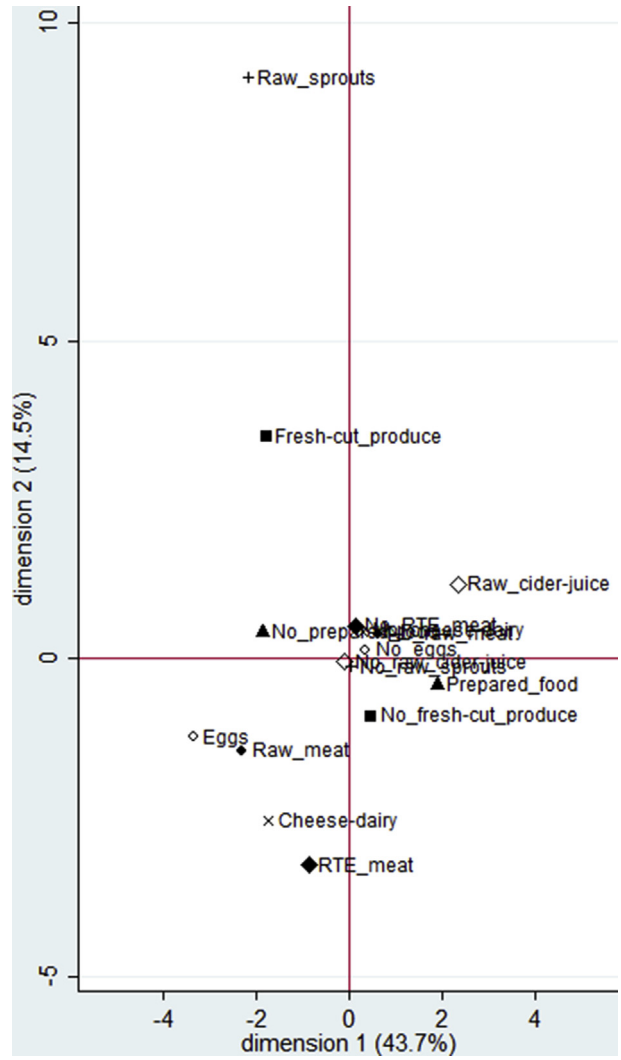

Fig. 1. Two-dimensional MCA plot of the relationship between types of TCS food products sold by farmers' market vendors in Ontario, Canada, in 2018.

Table 3
Behaviours requiring handwashing observed for 10 min, or up to 10 consecutive customer transactions, among 454 farmers' market vendors.

Behaviours	Mean frequency observed per vendor	SD	Range
Handle cash before handling food again	2.507	2.000	0–10
Touch mouth, hair, or face	0.797	0.896	0–5
Eat or drink	0.474	0.680	0–4
Handle phone	0.396	0.638	0–3
Handle a raw food before a ready-to-eat food	0.167	0.542	0–5
Handle other personal belongings	0.093	0.326	0–2
Handle waste container or garbage	0.046	0.230	0–2
Handle dirty dishes or utensils	0.024	0.224	0–4
Pick up something from the ground	0.024	0.168	0–2
Cough, sneeze or blow nose	0.011	0.104	0–1
Smoke	0.007	0.081	0–1
Touch animals	0.004	0.066	0–1
Total behaviours requiring handwashing	4.551	2.616	0–15

Table 4
Results of two multilevel mixed-effects logistic regression models evaluating vendors' use of two targeted food safety behaviours.

Behaviour outcome/covariate	OR	95% CI	*P* value
Use of cold-holding temperature control for all foods requiring it (yes vs. no)[a]			
Sells raw meat, poultry, or fish (yes vs. no)	5.361	2.823, 10.179	< 0.001
Sells ready-to-eat meat (yes vs. no)	0.136	0.059, 0.315	< 0.001
Sells fresh-cut leafy green vegetables, tomatoes, or melons (yes vs. no)	0.014	0.003, 0.062	< 0.001
Sells cheese or dairy products (yes vs. no)	0.169	0.077, 0.368	< 0.001
Use of bare hands to handle food on at least one occasion (yes vs. no)[b]			
Sells prepared foods (yes vs. no)	0.085	0.042, 0.171	< 0.001
Vendor engages in multiple roles (yes vs. no)	2.515	1.282, 4.932	0.007
Market operates on a weekend (yes vs. no)	2.098	1.260, 3.492	0.004

[a] N = 410 vendors, 57 markets. Overall model significance (Wald chi^2 test): $P < 0.0001$. Market-level variability: ICC = 0.015.
[b] N = 454 vendors, 59 markets. Overall model significance (Wald chi^2 test): $P < 0.0001$. Market-level variability: ICC < 0.0001.

behaviour requiring handwashing was handling cash before handling food again (mean frequency of 2.51 times per vendor, SD = 2.00). Overall, vendors were observed engaging in a mean of 4.55 behaviours requiring handwashing (SD = 2.62).

Only 13 vendors (2.9%) attempted to wash their hands during the observation period, each on only one occasion. None of the observed handwashing attempts was classified as adequate (i.e. washing with soap and water for ≥10 s). Two vendors (0.4%) used hand sanitizer during the observation period, one of which also washed their hands. During the observed transactions, most vendors handled food with their bare hands on at least one occasion (79%), while 46% used their bare hands for all instances of observed food contact. Other methods of handling food included tongs or other utensils (48%), hands with gloves (14%), and napkins or plastic bags (6%). Multiple methods were used by some vendors. Of the 65 vendors who wore gloves, only 25 (39%) changed them during the observation period, on either one (n = 23) or two (n = 2) occasions.

3.4. Regression modelling results

The results of the multilevel mixed-effects logistic regression models are shown in Table 4. For the model examining vendors' use of cold-holding temperature control methods for all foods requiring it, four statistically significant covariates were identified (Table 4). This behaviour was more commonly observed among vendors selling raw meats, poultry, and fish (odds ratio [OR] = 5.36; 95% confidence interval [CI] = 2.82, 10.12), and less commonly observed among vendors selling ready-to-eat meats, fresh-cut leafy green vegetables, tomatoes, and melons, and cheese and dairy products (Table 4). Only 1.5% of the residual variance in this outcome was explained by market-level (vs. vendor-level) variability.

Three statistically significant covariates were identified for the model examining vendors' use of bare hands to handle foods on at least one occasion (Table 4). Vendors who sold prepared foods were less likely to engage in this behaviour (OR = 0.08; 95% CI = 0.04, 0.17), while those who engaged in multiple roles (e.g. handling cash and food) were more likely (OR = 2.51; 95% CI = 1.28, 4.93) to handle food with their bare hands. In addition, this behaviour was more commonly observed at markets operating on the weekend (OR = 2.10; 95% CI = 1.26, 3.49) than those operating on weekdays. Market-level variability had no impact on the residual variance in this outcome (ICC < 0.0001). Univariable modelling results are reported as supplementary material.

4. Discussion

This study reports on the first comprehensive assessment of food safety practices at farmers' markets in the Greater Toronto Area of Ontario, Canada. We found a wide range of market sizes in the study area, with approximately half of vendors selling one or more TCS food products. In terms of available infrastructure, most markets had at least one communal handwashing sink, but < 50% had visible electric access that could be used by vendors (e.g. for powering refrigeration or hot-holding units). Handwashing sinks were observed more often than has been noted in previous studies of farmers' markets (Young et al., 2017), while the availability of electrical access has also been reported to be low in other studies (McIntyre et al., 2014; Pollard et al., 2016). Nearly all markets appeared to allow pets on site, with only one indoor market prohibiting them from entering, which also corresponds with previous research (Young et al., 2017). Given that numerous enteric disease illnesses and outbreaks have occurred due to direct and indirect animal contact (e.g. at petting zoos and related venues) (Conrad, Stanford, Narvaez-Bravo, Callaway, & McAllister, 2017), the availability and promotion of handwashing areas and hand sanitizers should be emphasized at farmers' markets where pets and other animals are allowed.

Most of the observed vendors were selling prepared foods, which is a broad category representing products for immediate consumption (e.g. composite meals, hamburgers, sandwiches), as well as other foods for consumption at home (e.g. soup stocks, dips and spreads, fermented foods such as tofu). As shown in Fig. 1, these vendors tended to be distinct from vendors selling other TCS foods. The four groups of vendors identified through MCA can be used to inform and target future food safety education and outreach efforts towards those who sell similar types of TCS food products. Given this was an observation study with no vendor interaction, we were not able to confirm what proportion of vendors were primary producers selling their own agricultural products, though a previous survey in Ontario estimated this percentage to be > 75% (Farmers' Markets Ontario, 2009).

While the appearance of general cleanliness and sanitation was very high among vendors observed in this study, approximately one-quarter of vendors did not use any methods of protecting at least some of their TCS foods from cross-contamination, which has also been identified as a behavioural gap among market vendors in previous studies (McIntyre et al., 2014; Scheinberg et al., 2018). Protection of food from cross-contamination is important to prevent transfer of foodborne pathogens via different vehicles (e.g. utensils, surfaces, hands) and environmental sources (e.g. dust, aerosols) (Todd, Greig, Bartleson, & Michaels, 2009). Also concerning is that > 50% of vendors who offered food samples did not provide single-serving utensils, which could also lead to cross-contamination. Future education with market vendors should emphasize the importance of cross-contamination prevention for TCS foods and food samples.

We identified gaps in vendors' use of cold- and hot-holding methods to maintain appropriate temperature control of TCS foods. Previous studies have also identified temperature control issues among farmers' market vendors (McIntyre et al., 2014; Pollard et al., 2016; Teng et al., 2004). The regression model highlighted that vendors selling ready-to-eat meats, fresh-cut leafy green vegetables, tomatoes, and melons, and cheese and dairy products were less likely to use cold-holding temperature control methods. Ready-to-eat meats, such as fermented

sausages, have traditionally been considered a safe product, but have caused numerous outbreaks of verotoxin-producing *Escherichia coli* in recent decades (Holck et al., 2011). If appropriate pathogen reduction measures are not taken during the manufacturing of these products, they could present food safety risks to consumers. Health inspection authorities should confirm the final a_w and pH of such products with vendors to determine their need for temperature control (U.S. Food and Drug Administration, 2017). We were not able to evaluate the intrinsic parameters of these products during this study, so it is likely that several of the ready-to-eat meats observed would not require temperature control due to their a_w, pH, and other factors (e.g. high salt levels).

The lack of cold-holding temperature control among vendors selling fresh-cut leafy green vegetables, tomatoes, and melons is not surprising, because the current food safety guidance manual for vendors in Ontario classifies all fruits and vegetables as "low risk" foods (Farmers' Markets Ontario, 2019b). Similarly, fresh fruits and vegetables are classified as "not potentially hazardous" in the public health inspector guide to farmers' markets in Ontario (ASPHIO, 2012). However, the U.S. Food Code recognizes fresh-cut leafy green vegetables, tomatoes, and melons as requiring temperature control, because the act of cutting exposes internal nutrients in these products that provide suitable conditions for microbial growth (Francis et al., 2012; U.S. Food and Drug Administration, 2017). In addition, numerous foodborne disease outbreaks have been linked to these products in recent years (Bennett, Littrell, Hill, Mahovic, & Behravesh, 2015; Callejón et al., 2015; Walsh, Bennett, Mahovic, & Gould, 2014). Additional outreach is needed with market vendors and health inspection authorities about possible food safety risks from these products and the need for temperature control. The lack of cold holding among cheese and dairy vendors is also a concern, as previous foodborne disease outbreaks have been linked to the sale of these products at farmers' markets (Young et al., 2017).

The most commonly observed behaviour requiring handwashing among vendors was handling money in between handling food, a practice that has also been reported at a high frequency in previous studies (Behnke & Seo, 2015; McIntyre et al., 2014; Scheinberg et al., 2018). Despite observing > 1300 total transactions among 454 vendors, we found that less than 3% of vendors attempted handwashing, which corresponds with a previous systematic review that found an average prevalence of handwashing among vendors of 4% across five studies (Young et al., 2017). This finding is not surprising given that handwashing stations were not visible at > 60% of individual vendor stalls, which could be related to a lack of financial capital among farmers' market vendors to invest in such infrastructure, or a lack of a requirement to have one available (Harrison, Critzer, & Harrison, 2016; Scheinberg et al., 2018). The low rate of handwashing may also be related to poor placement of the handwashing stations (e.g. away from the customer display areas), impractical and inconvenient station setups (e.g. most stations required vendors to hold a waterspout to wash their hands), and a lack of perceived risk from not handwashing. Enhanced education, training, and additional policy strategies should be provided at farmers' markets in Ontario to facilitate hand hygiene among vendors selling TCS foods.

The availability and use of hand sanitizer was also found to be very low in this study. The current food safety guidelines for farmers' market vendors in Ontario recommend that hand sanitizer only be used as an alternative to handwashing when soap and water are not available and when only fruits, vegetables, or pre-packaged foods are sold (Farmers' Markets Ontario, 2019b). In the food service industry, handwashing is recommended as a more reliable method of ensuring hand hygiene than hand sanitizers, as previous research has found that hand sanitizers may have limited efficacy against some pathogens (e.g. norovirus) and may not be as effective as handwashing when hands are visibly soiled (Foddai, Grant, & Dean, 2016; Todd et al., 2010). However, recent studies have found that a two-step application technique may be of comparable efficacy to handwashing in agricultural settings (de Aceituno et al., 2015, 2016). Additional research is warranted on the possible effectiveness, utility, and practicality of hand sanitizers in farmers' market settings.

A high proportion of vendors in this study engaged in bare hand contact of TCS foods during the observation period, which corresponds with findings from a recent study in Pennsylvania, U.S. (Scheinberg et al., 2018). The regression modelling of this behaviour identified that it was less commonly practised among vendors who sold prepared foods, and more commonly practised among those who engaged in multiple work roles at the vendor stall (Table 4). Behnke and Seo (2015) also found that farmers' market vendors in Indiana who engaged in multiple work roles demonstrated a higher frequency of handwashing violations. When more than one employee is working at a vendor stall, duties should ideally be segregated to prevent possible cross-contamination. A higher likelihood of bare hand contact with food was also observed at weekend vs. weekday markets, which could be related to higher customer activity during these days. We did not capture direct information about the number of customers at each market, nor is that information available publicly, but we informally observed higher customer activity at weekend-operating markets. In busier periods, vendors may be less likely to engage in food safety practices, as has been noted among other food service workers (Clayton, Clegg Smith, Neff, Pollack, & Ensminger, 2015). Future education and training on avoiding bare hand contact should be emphasized among market vendors along with the importance of proper hand hygiene.

There are some limitations of this study. Although interrater reliability for the observational instrument was assessed by two researchers in a pre-test and found to have excellent overall agreement, only one researcher assessed all outcomes in the main study. Therefore, it is possible that some misclassification errors in the observed vendor practices could have occurred. Although the technician was trained to be as discrete as possible, some vendors could have noticed that they were being observed and altered their practices. However, the technician was never approached by any vendors to inquire about this possibility. Given that we visited all farmers' markets identified in the study area, we believe the results have strong internal validity. However, we visited each market and observed each vendor on only one occasion, so the practices observed reflect their food safety behaviours at one point in time and may not be representative of their usual or routine practices. In addition, the study setting was a concentrated urban and suburban area, and results may be different for farmers' markets in other (e.g. rural) areas in Ontario.

Because vendors were not interviewed or surveyed directly in this study, we were not able to capture information on sociodemographic factors that could not be observed directly (e.g. age, education, sales information, previous food safety training). In addition, we were not able to explore other possible determinants of vendors' food safety behaviours, including their knowledge, attitudes, and other psychosocial factors. These characteristics are known to be related to the food safety practices of food handlers in other settings (Young, Thaivalappil, Waddell, Meldrum, & Greig, 2018). Future studies on this topic should consider incorporating a vendor survey or interviews to explore additional sociodemographic, psychosocial, and policy factors as possible determinants of vendor behaviours.

Public health and food safety authorities should work collaboratively with the farmers' market industry to identify pragmatic strategies and opportunities to enhance the use of food safety practices among market vendors. A previous study in Florida found that few farmers' market vendors reported receiving prior food safety training, but most were interested in learning more about food safety (Simonne, Swisher, & Saunders-Ferguson, 2006). Therefore, specialized food safety training programs for farmers' market vendors should be developed, including both in-person and online offerings, which exist in other jurisdictions (McIntyre et al., 2014). In addition, regular on-site assessments should be conducted by public health inspectors to promote and encourage food safety.

5. Conclusion

This study has identified some important gaps in the use of recommended food safety behaviours among farmers' market vendors, namely in the use of methods to protect TCS foods and food samples from cross-contamination, storing foods using adequate hot- and cold-holding temperature control methods, and practicing appropriate hand hygiene. These areas should be prioritized in future education, training, and outreach initiatives with farmers' markets vendors. Market managers also have an important role in setting food safety policies and promoting food safety among vendors. However, given that market-level variability was not an important contributor to variation in vendors' use of two targeted behaviours (cold-holding temperature control and bare hand contact with TCS foods), future outreach efforts to improve these behaviours should primarily target individual vendors rather than market managers only. Opportunities for future research could include further investigation of the sociodemographic and psychosocial determinants of vendor behaviours, the viability and utility of alternative methods to traditional handwashing in a market setting, and the effectiveness of different education and training approaches for market vendors.

Conflicts of interest

Declarations of interest: none.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodcont.2019.106875.

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